INCH-POUND

A-A-60020
March 31, 2021

COMMERCIAL ITEM DESCRIPTION

SOCKS, COLD WEATHER SYSTEM OF

1. SCOPE. This Commercial Item Description (CID) covers four (4) types of wool socks intended for wear in cold and extreme cold weather environments by personnel of the Department of Defense.

2. CLASSIFICATION. Types I, II, III and IV socks shall be available in the following types and sizes:

2.1 Types.

Type I - Liner
Type II - Light Weight Sock
Type III - Mid Weight Sock
Type IV - Heavy Weight Sock

2.2 Sizes. The sizes shall be available in the following sizes.

SIZE SCHEDULE:

Shoe Size	X-Small	Small	Medium	Large	X-Large
Men	2.5-5	5.5-7.5	8-9.5	10-12	12.5-14.5
Women	4.5-7	7.5-9.5	10-11.5	12-14	----

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AMSC N/A FSC 8415

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3. SALIENT CHARACTERISTICS.

3.1 Basic materials, (Types I, II, III, and IV). The socks shall contain the following minimum percentages of wool that is 22.5 micron or finer, Nylon and Lycra® elastane (see Table I).

TABLE I. Fiber composition requirements for Types I, II, III, and IV.

Type/Fiber Percentage % (minimum for each fiber) 1/	Wool (22.5 microns or finer) (%) min.	Nylon (%) min.	Lycra® Elastane (%) min.
Type I - Liner	60%	10%	2%
Type II – Light Weight	65%	10%	2%
Type III – Mid Weight	70%	15%	2%
Type IV – Heavy Weight	75%	13%	2%

1/ Total overall fiber content may vary by (± 3)% from the labeled content as long as final fiber content adds up to 100 percent

3.2 Construction. All Sock types shall be mid-calf in length, knit with arch support, and seamless toe construction or recessed conventional seam. All types shall have double covered elastic yarn that is continuous in every course of the welt, cuff and leg, after the makeup, from the interior beginning of the welt, through the leg and including the arch support area. Socks shall have 2-ply nylon impact zone reinforcements.

3.2.1 Type I. Type I, Sock liner shall be constructed as a flat jersey knit.

3.2.2 Type II. Type II, Light weight socks shall have terry insulation/cushioning in the entire heel, toe, and footbed area. The minimum terry loop depth shall be 5mm inside diameter. Ventilation panels shall be knit into the top of foot area.

3.2.3 Type III. Type III, Mid weight socks shall have a terry loop cushioning in the toe and heel. The minimum terry loop depth shall be 5mm inside diameter.

3.2.4 Type IV. Types IV, Heavy weight socks shall have a terry loop cushioning in the toe and heel. The minimum terry loop depth shall be 5mm inside diameter.

3.3 Color (all types). Color should shall be Black or Coyote Brown 498.

3.3.1 Visual shade matching. The color and appearance of the socks specified in 3.3 shall match the standard sample when viewed using the AATCC EP 9, Option C (see 7.3.1), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500K (± 200) illumination of 100 (±20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

3.4 Performance requirements. The finished socks shall conform to the requirements in Table II when tested according to the test methods in Table II.

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TABLE II. Material performance requirements for Types I, II, III, and IV.

Characteristic	Requirement				Test Method
Fiber Analysis, percent	All Types See 3.1				AATCC TM20A
Bursting strength, pounds (min.) Toe Heel Leg	Type I 60 60 60	Type II 70 60 60	Type III 70 60 60	Type IV 90 90 90	ASTM D3787
Dimensional stability, percent (max.) Foot Length -FL Calf Length CL Foot Width-FW Calf Width-CW	All Types 7.0% max average of FL, CL, FW and CW				AATCC TM150 (3X)1/
Wicking, mm/s (30 minutes) (min.)	Type I 0.020	Type II 0.020	Type III 0.030	Type IV 0.015	AATCC TM197, 30 minutes
Stretch and Recovery, Percent Calf Stretch (min.) Calf Growth (max.) Foot Stretch (min.) Foot Growth (max.)	Type I 50 20 50 20	Type II 50 20 50 20	Type III 40 20 40 20	Type IV 30 15 30 15	ASTM D6614, Hold at 4 lbs. for 5 minutes
Full Conditioned Sock, Weight (grams) (single sock, size Large) Tolerance	Type I ≤ 42 g +0,-3%	Type II ≤ 50 g +0,-3%	Type III ≤ 60 g +0,-3%	Type IV ≥ 63 g -0,+3%	Condition at least 4 hours at 70°F (± 2° F) and 65% (±2%) relative humidity

1/ 3 cycles, machine cycle 1, wash temp IV, tumble dry A-iii

3.5 Finished measurements. The finished measurements shall be in accordance with Tables III through VI.

TABLE III. Finished measurements, Type I (inches).

Location		X-Small	Small	Medium	Large	X-Large	Tolerance
A	Leg Length	11.25	12	12.5	13	13.75	± 0.5
B	Double Welt Top	2	2	2	2	2	± 0.25
C	Foot Length	7.5	8	9	9.5	10	± 0.5
D	Foot Width	3	3.25	3.25	3.5	3.5	± 0.25

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TABLE IV. Finished measurements, Type II (inches).

Location		X-Small	Small	Medium	Large	X-Large	Tolerance
A	Leg Length	9.5	10	10.75	11.25	12	± 0.5
B	Double Welt Top	1.5	1.5	1.5	1.5	1.5	± 0.25
C	Foot Length	8	8.5	9	9.5	10	± 0.5
D	Foot Width	3.25	3.5	3.5	3.75	3.75	± 0.25

TABLE V. Finished measurements, Type III (inches).

Location		X-Small	Small	Medium	Large	X-Large	Tolerance
A	Leg Length	8	9.75	10.75	11.5	13	± 0.5
B	Double Welt Top	1.5	1.5	1.5	1.5	1.5	± 0.25
C	Foot Length	8.5	9	9.5	10	10.5	± 0.5
D	Foot Width	3	3.5	3.5	4	4	± 0.25

TABLE VI. Finished measurements, Type IV (inches).

Location		X-Small	Small	Medium	Large	X-Large	Tolerance
A	Leg Length	8	9.5	11	12	13	± 0.5
B	Double Welt Top	1.5	1.5	1.5	1.5	1.5	± 0.25
C	Foot Length	7.5	8	9	9.5	10	± 0.5
D	Foot Width	3.25	3.75	4	4	4	± 0.25

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Method of measuring for Tables III, IV, V and IV (see Figure 1):
(A) Leg length - With the ruler running parallel to the ribbing in the leg, the Leg length is measured from very top of sock to bottom of the heel at the transition point (2).
(B) Double welt top - The length of the doubled fabric area in the top of the sock.
(C) Foot length - With the ruler running parallel to the wale direction in the foot, the Foot length is measured from the top of the heel at the transition point (1) to the center end of the toe.
(D) Foot width - With the ruler placed on top of the gore line (3) which runs through the center of the heel, the Foot width is measured from the back of the heel to the top of foot.

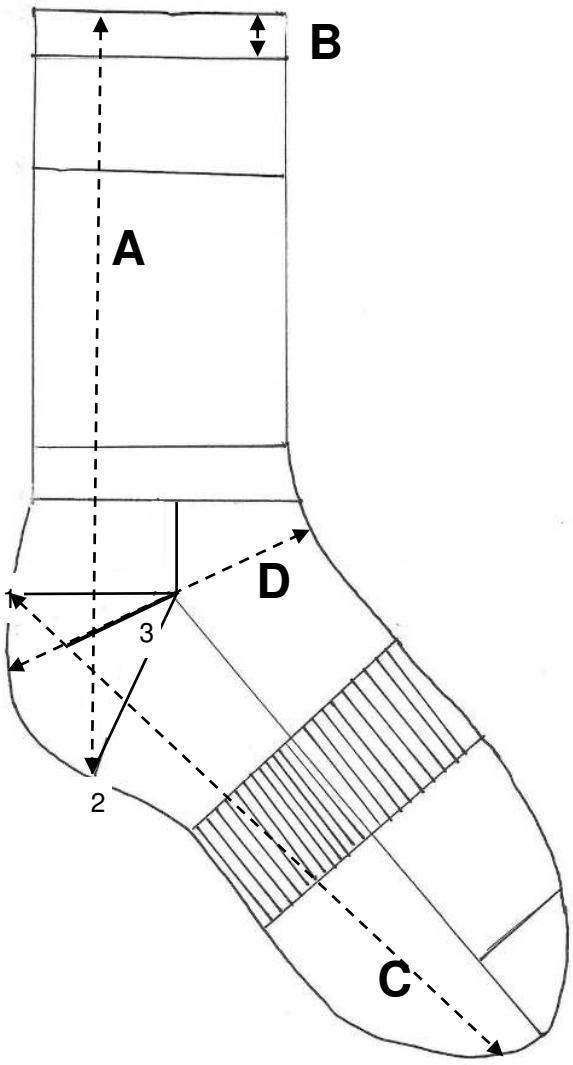


FIGURE 1. Method of measuring.

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3.6 Label/marking. The socks shall be packaged as a pair with a commercial type label containing the contractor's name, style, size, fiber content breakdown, wash/care instructions, and UPC number.

3.7 Toxicity. The finished material shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 5.6. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.8 Workmanship. The socks shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality limits.

4. REGULATORY REQUIREMENTS. Unless otherwise specified the offer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The finished socks provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Market acceptance. This finished socks must have been sold to the commercial market or the Government.

5.3 Visual examination. The finished socks shall be examined for the defects listed below as defined in ASTM D3990:

5.3.1 Defects. Any hole, runs, dropped stitch (including holes looping or seaming), cut, or tear; color not as specified; any part shaded or noticeable variation in size; non-uniform in overall appearance, not well shaped, slubby yarn, stain, shade bar or spot, rancid or bad odor, any burn or scorch, skipped stitches, bulky, run-offs, or broken thread; any dimension or weight not as specified, label missing, incorrect, misplaced or illegible; measurement of item not as specified, barcode or UPC code omitted or not readable by a scanner (human readable interpretation (HRI)), omitted or illegible; bar code miss code, not visible on packaged item, barcode causes damage to the item; any item not packaged in accordance with contract or purchase order.

5.4 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order (see 7.6).

5.5 End item dimensional examination. The Socks shall be examined to verify conformance to the finished measurements in 3.5. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of socks. The sample unit shall be one (1) pair of socks. The inspection level, expressed in terms of defects per hundred units, shall be as specified in the contract or purchase order.

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5.6 Toxicity test. When required (see 7.6), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of the studies indicate the socks is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 7.2.4). If the toxicity requirement (see 3.7) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or purchase order (see 7.6).

7. NOTES

7.1 Sources of Government documents.

7.1.1 Copies of Government documents are available online at https://quicksearch.dla.mil.

7.1.2 Wool Products Labeling Act. (Applications for copies should be addressed to Textile Section, Division of Enforcement, Federal Trade Commission, Washington DC 20580.)

7.2 Source of non-Government documents.

7.2.1 AATCC Test Methods are available online at https://www.aatcc.org.

7.2.2 ASTM Test Methods are available online at https://www.astm.org.

7.2.3 ASQ/ANSI Z1.4 – Sampling Procedures and Tables for Inspection of Attributes are available online at https://asq.org.

7.2.4 Repeat Insult Patch Test - Modified Draize Procedure -Principles and Methods of Toxicology, A Wallace Hayes (editor) are available online at https://www.crcpress.com.

7.3 Standard samples. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

7.3.1 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be “The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.”

7.4 Intended use. These socks are intended for wear by male and female military personnel of the Department of Defense operating in cold and extreme cold weather environments.

7.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

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7.6 Ordering data. The contract or purchase order should specify the following:

a. Title, number, and date of this Commercial Item Description.
b. Size required (see 2.1).
c. Labeling information (see 3.5).
d. When toxicity testing is required (see 3.7 and 5.6).
e Regulatory requirements (see 4).
f. Product conformance provisions (see 5).
g. Acceptance criteria (see 5.4).
h. Packaging requirement (see 6).

7.7 Known suppliers.

Crescent Inc.
Omni Wool Tactical
Crescent Sock Company
527 E Willson St
Niota, TN 37826
Wood Talkington
wtalkington@crescent-inc.com

Darn Tough
364 Whetstone Drive
PO Box 307
Northfield, VT 05663
Shannon Shrader
sshrader@darntough.com

Fits
military@fitssock.com

Fox River
227 Poplar St.
Osage, Iowa 50461
Adam Rotruck
Email: arotruck@govtools.com

Legend Compression Wear
PO Box 326
Denton, NC 27239
John J. Thomas
Email: john@legendcompressionwear.com

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Nester Socks
1546 Carter Street
Mount Airy, NC 27030
United States
Kerry Nester
kerry@nesterhosiery.com

Serket USA
Black Canyon Concepts LLC dba Serket USA
711 Commerce Way Suite 8
Jupiter, FL 33458
Randy Webb
Randy@SerketUSA.com

7.8 Subject term (key word) listing.

Footwear
Wool

MILITARY INTERESTS:

Custodians
Army-GL
Air Force-11

CIVIL AGENCY COORDINATING ACTIVITY: GSA - FAS

PREPARING ACTIVITY:
DLA -CT

(Project: 8415-2021-028)

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